Chapter 1: PI feedback controller for active steering

This second part uses the same BMW 730 lateral model to design an active steering controller. The measured output is the yaw rate. The main objective is constant yaw-rate tracking and rejection of slowly varying steering or road disturbances, while keeping the closed-loop response at least as fast as the open-loop vehicle dynamics.

$$C_PI(s) = Kp + Ki/s = (Kp\ s + Ki)/s$$

Since the steering-to-yaw-rate transfer function is stable, minimum phase and of relative degree one in forward motion, a PI controller is sufficient to achieve zero steady-state error for a constant reference and constant matched disturbances. The closed-loop characteristic polynomial is obtained from the Diophantine equation a(s)s + b(s)(Kp s + Ki).

$$lambda(s) = s\text{^}3 + (A1 + B1\ Kp)s\text{^}2 + (A2 + B1\ Ki + B2\ Kp)s + B2\ Ki$$

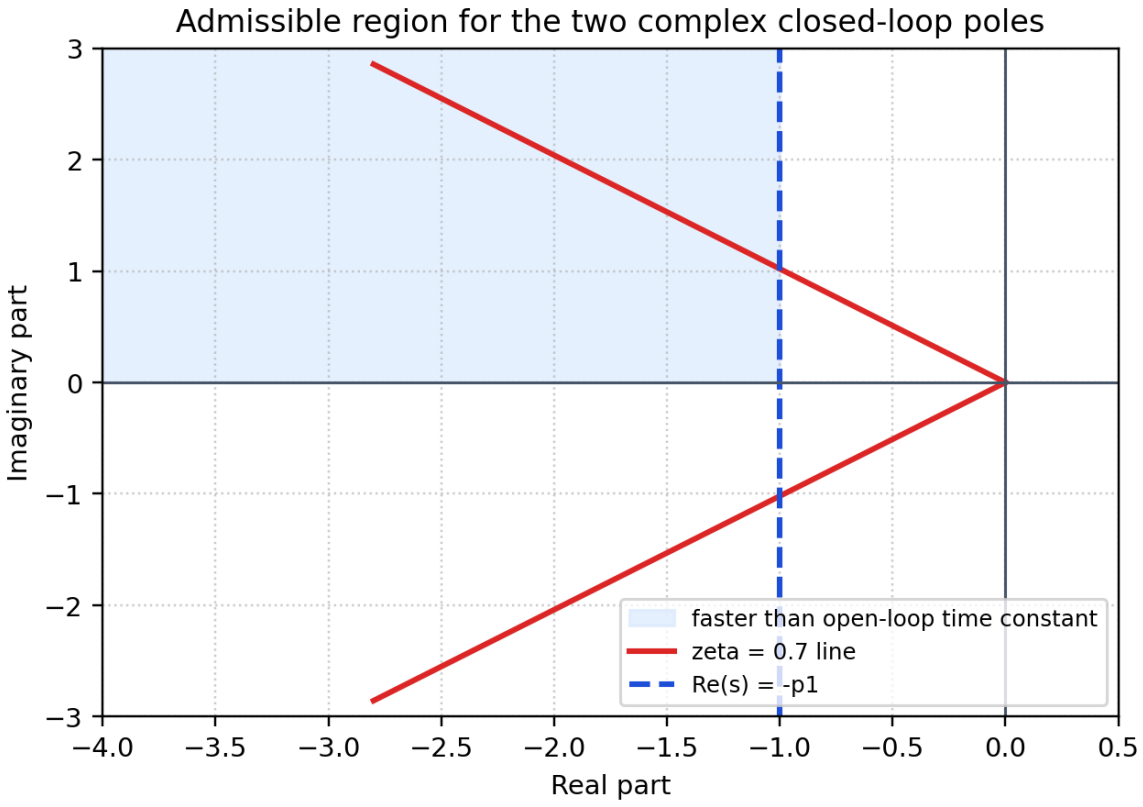


Figure 1. Region used for selecting the two complex closed-loop poles.

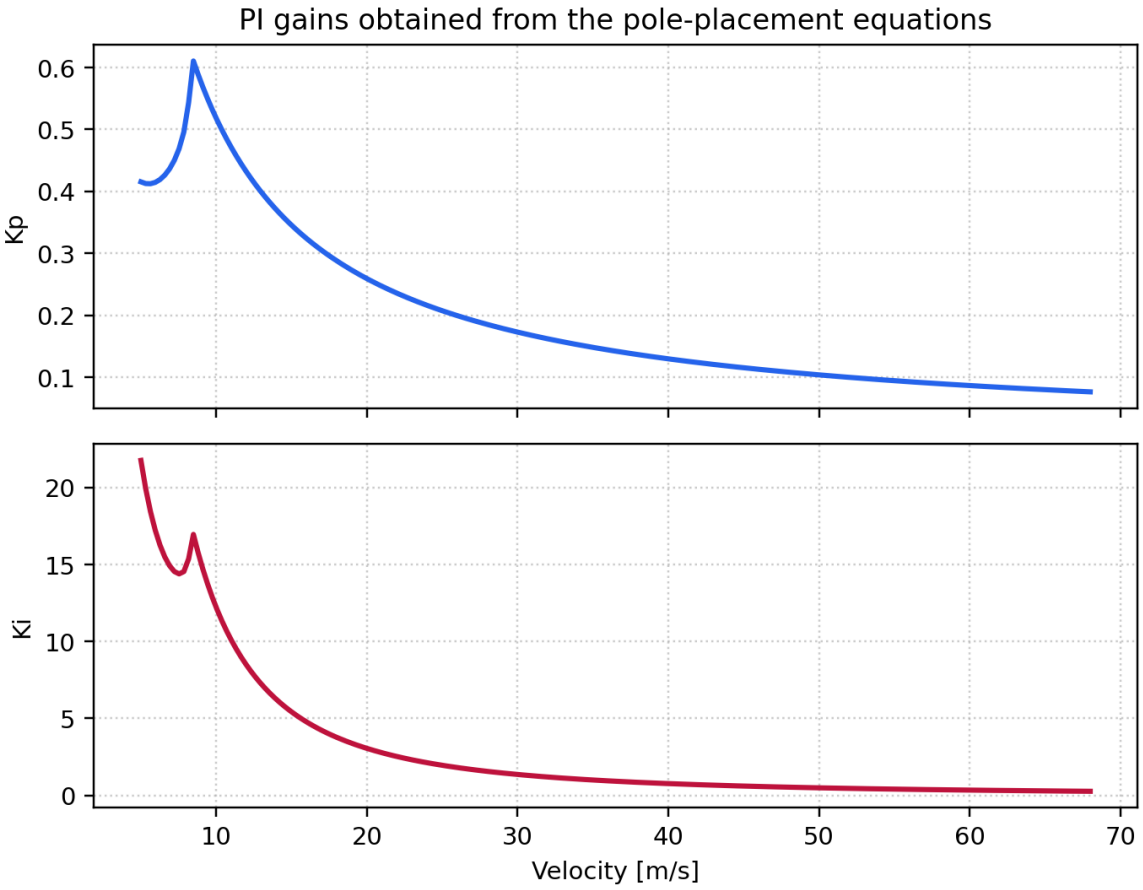


Figure 2. Computed PI gains as functions of the forward velocity.

Chapter 2: Luenberger observer and Wiener filter

The yaw-rate sensor does not directly provide the full state x = [v_y, omega]^T. Therefore, an observer is required when state estimates are used inside the feedback loop. Two estimation strategies are considered: deterministic pole-placement using a Luenberger observer and stochastic filtering using a Wiener/Kalman structure.

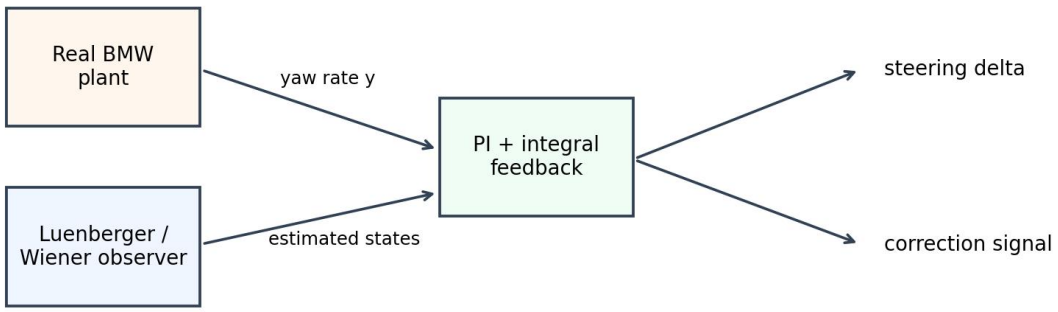


Figure 3. Observer-based active steering structure.

2.1 Luenberger observer

$$\text{xhat_dot} = A\,\text{xhat} + B\,\text{delta} + L(y - C\,\text{xhat})$$

The observer gain L is chosen by applying pole placement to the dual pair (A^T, C^T). In the simulations the observer poles are placed faster than the open-loop vehicle poles so that the estimation transient does not dominate the closed-loop steering response.

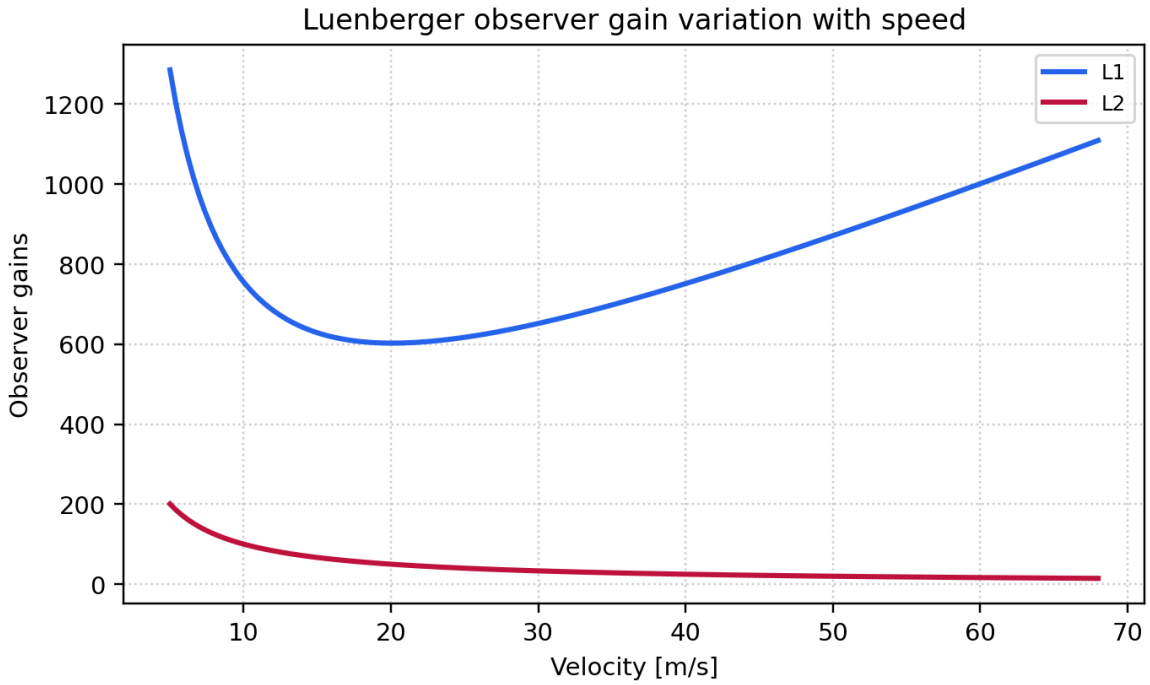


Figure 4. Luenberger observer gain variation with vehicle speed.

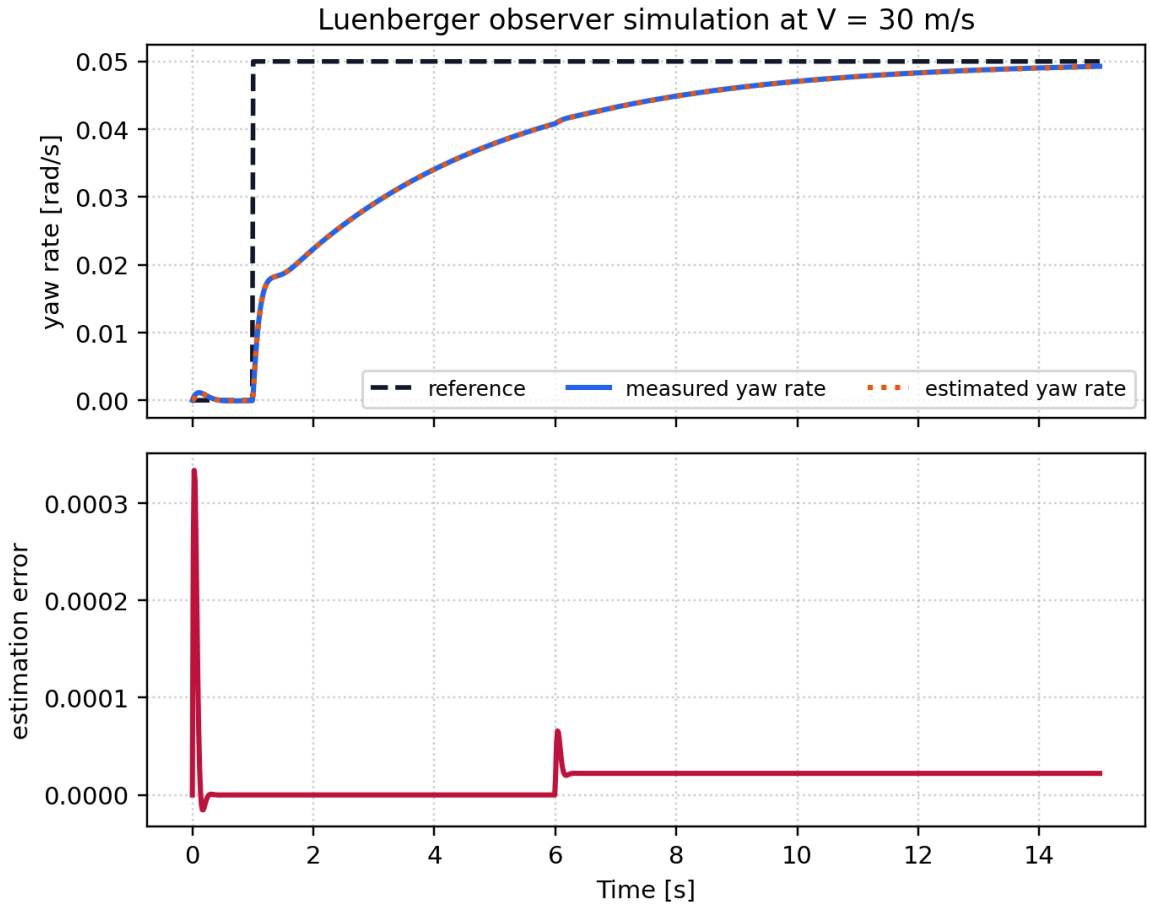


Figure 5. Luenberger observer tracking and estimation error.

2.2 Wiener / Kalman filter

For the Wiener filter, process and measurement noises are introduced and the gain is obtained from the continuous-time Riccati equation. This design balances confidence in the model and confidence in the sensor. Compared with a pure pole-placement observer, the Wiener filter gives a systematic way to tune the estimator in the presence of noisy yaw-rate measurements.

$$A^T P + P A - P C^T R^{-1} C P + Q = 0, \quad K_w = P C^T R^{-1}$$

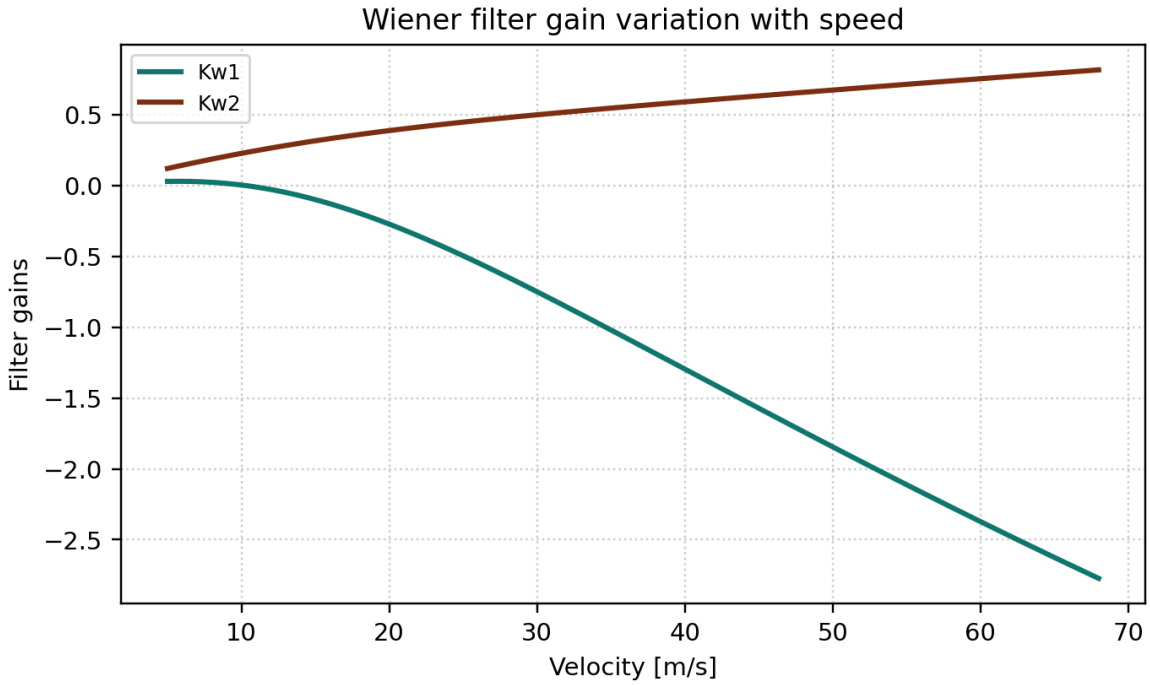


Figure 6. Wiener filter gain variation with forward speed.

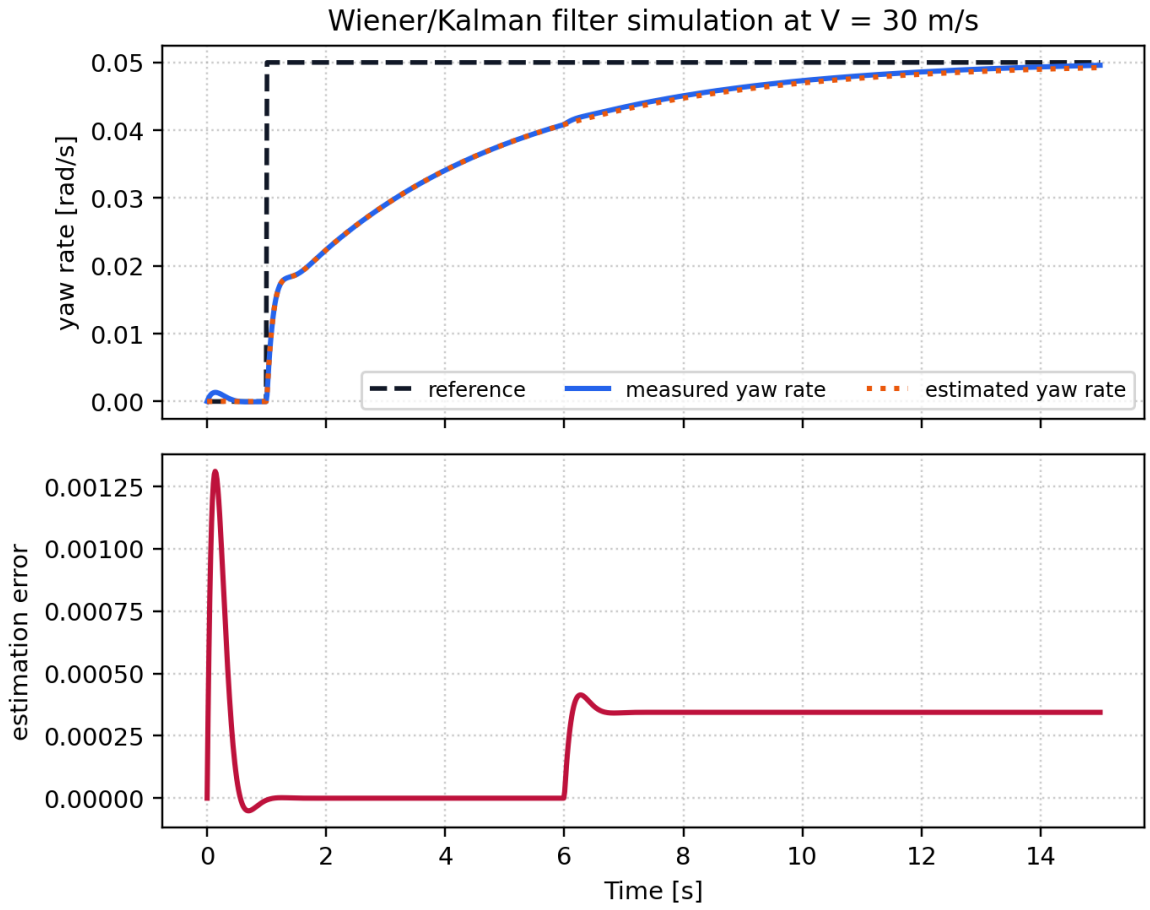


Figure 7. Wiener/Kalman filter tracking and estimation error.

Chapter 3: Observer-based controller with integral action

The final controller combines the PI integral state with the estimated yaw-rate state. The integral term provides steady-state accuracy, while the observer supplies the missing internal state. The steering command is calculated from the yaw-rate tracking error and the integral of this error.

$$\text{xi_dot} = \text{omega_ref} - \text{omega_hat}, \qquad \text{delta} = \text{Kp}(\text{omega_ref} - \text{omega_hat}) + \text{Ki}\,\text{xi}$$

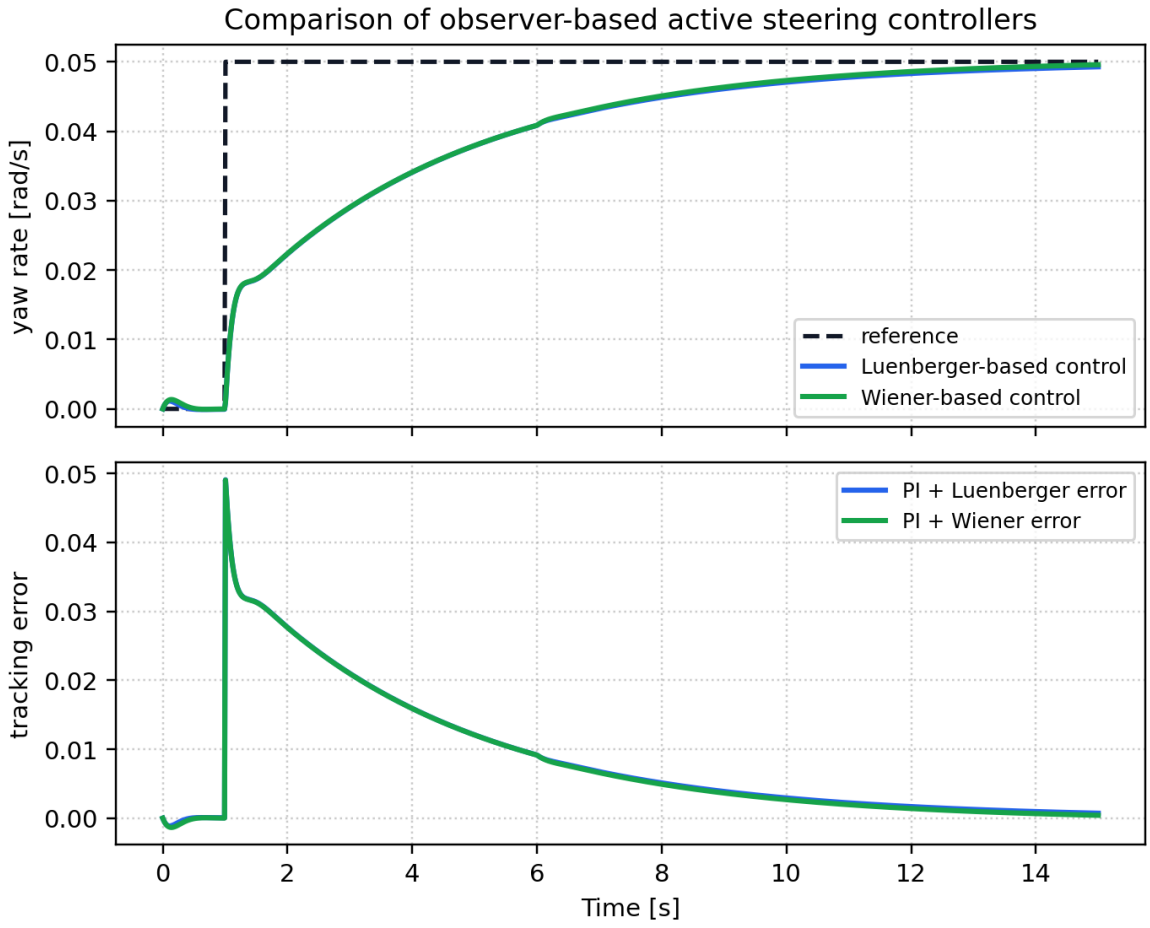


Figure 8. Closed-loop comparison between Luenberger-based and Wiener-based active steering.

Chapter 4: Discussion and comparison

Both observer-based solutions achieve yaw-rate regulation with zero steady-state error for the constant reference. The Luenberger observer is easier to design and gives a predictable convergence speed through pole assignment. The Wiener filter is more useful when the yaw-rate measurement is noisy, because the estimator gain is obtained from the assumed noise covariance matrices. In both cases the integral action remains essential for disturbance rejection.

Controller	Main advantage	Main limitation
PI only	Simple, direct yaw-rate tracking	Requires measured output only; no full-state estimate
PI + Luenberger	Fast deterministic state reconstruction	Gain depends on selected observer poles
PI + Wiener filter	Better noise-aware estimation	Requires Q and R covariance tuning

Conclusion

The active steering controller can be designed around the yaw-rate transfer function because it is stable and minimum phase in forward driving. The PI controller handles constant tracking and disturbance rejection. Adding a Luenberger observer or Wiener filter allows the same controller to operate using estimated internal states, and the Wiener solution is preferable when measurement noise is important.